FeedFlipNets: Deterministic Direct Feedback Alignment with Ternary Forwards–Theory, Evidence, and a Reproducible Baseline*

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Abstract

Training without exact weight-transport symmetry is attractive for specialized hardware, simulators, and as a biologically plausible mechanism, yet empirical accounts of *Direct Feedback Alignment* (DFA) remain scattered and difficult to audit. We present **FeedFlipNets**, a deterministic baseline that couples DFA with *ternary forward* weights and high-precision *shadow* parameters. The evaluation follows four principles—determinism (fixed seeds and scripted reports), coverage (eight canonical datasets across four modalities in offline and real modes), fairness (matched model families, budgets, and pre-specified hyperparameter grids), and deployability (sparsity, throughput, and CPU latency alongside accuracy).

On compact benchmarks, DFA (float or ternary forwards) tracks backpropagation on MNIST and UCR GunPoint and is statistically tied on AG News. On 20 Newsgroups, DFA (float) exceeds the matched backprop baseline under our configuration, with the gap most apparent at intermediate learning rates. Ternary forwards yield zero ratios between 0.5 and 0.75 with competitive throughput, exposing accuracy–sparsity Pareto fronts and favorable memory footprints.

We analyze directional agreement via a sign-match statistic p, typically above $\frac{1}{2}$, and prove finite-time convergence to a stationary neighborhood under ternary noise whenever $p > \frac{1}{2}$. Orthogonal or Hadamard feedback raises the alignment floor and stabilizes deeper stacks without sacrificing speed. Collectively, the results provide a transparent, reproducible account of when ternary-forward DFA *does* and *does not* work. Scope is intentionally limited to shallow MLPs and the released configurations; code, seeds, and scripts reproduce figures and tables end-to-end.

Keywords: Direct Feedback Alignment, ternary networks, quantized training, structured feedback, alignment dynamics, determinism, deployability, reproducibility.

Headline	Metric	Value
Vision (MNIST, real)	Test Acc. (BP vs DFA float)	96.46% ± 0.03% vs 95.14% ± 0.08%
Text (20NG, real)	Test Acc. (BP vs DFA float)	.199 ± .013 vs .321 ± .005
AG News (real)	Test Acc. (BP vs DFA float)	90.05% ± 0.07% vs 89.98% ± 0.05%
Time-series (UCR)	Best Test Acc. (Ternary DFA, real)	100.00%
Sparsity (Ternary)	Zero ratio (typical)	≈ 0.5–0.75
Determinism	Seeds, fixtures	Yes

Key Metrics (condensed). Representative values from our benchmark suite.

Reproducibility Statement: see Appendix.

Abbreviations: 20NG = 20 Newsgroups; Cal. = California; BoW = bag-of-words; ps = per_step; pe = per_epoch.

*Artifact availability: code, data splits, and scripts at `https://github.com/akigogikar/FeedFlipNets`. Commit: cc8103b7d7a7.

1 Introduction

Training without weight-transport symmetry is compelling for hardware, simulators, and biological plausibility, but progress has been slowed by fragile experiments and inconsistent claims. Conventional backpropagation attains strong accuracy yet relies on exact transpose transport in the backward pass. *Direct Feedback Alignment* (DFA) [18, 20] removes that requirement by broadcasting output errors through fixed random matrices, while low-precision networks [7, 23, 15, 24] reduce compute and memory via quantization. What the field lacks is a *credible, reproducible baseline* unifying these ideas: a clear picture of when DFA with ternary forward weights succeeds, where it fails, and what trade-offs it entails on familiar, compact benchmarks.

We provide that baseline with **FeedFlipNets**[1]: a fully deterministic pipeline that pairs DFA with *ternary forward* weights and high-precision *shadow* parameters. Runs are scripted end-to-end, seeds are fixed, confidence intervals are reported, and all tables/figures are auto-generated from logs—so claims can be audited and reproduced on a single workstation. Across vision, time-series, tabular, and text tasks, we map accuracy–sparsity–throughput Pareto fronts, track alignment statistics that connect empirics to theory, and evaluate structured feedback (orthogonal/Hadamard) to stabilize training in deeper stacks.

Contributions.

- **A deterministic, theory-anchored reference for DFA with ternary forwards.** We release a minimal, scripted stack with fixed seeds, deterministic data splits, logging, and visualization—intended as a trustworthy baseline for feedback-aligned and low-precision learning.
- **Finite-time analysis tied to measurable signals.** We formalize the update, prove a stationarity bound under ternary quantization noise using a sign-advantage assumption, and give an alignment lemma for structured feedback; the required proxies (sign-match p, cosine alignment ρ) are logged during training.
- **A careful empirical map of trade-offs.** On compact benchmarks, DFA (float or ternary forwards) closely tracks backpropagation on vision and time-series, ties on some text tasks, and exposes clear accuracy–sparsity–throughput Pareto fronts; structured feedback narrows variance and raises an alignment floor.
- **Practical guidance for deployability.** We distill design rules for feedback structure, flip schedules, thresholds, and clipping that yield stable training and sparse, hardware-friendly models.

Together, these results replace anecdote with a transparent, reproducible account of when ternary-forward DFA *does* and *does not* work. Scope is intentionally narrow: small multilayer perceptrons on standard datasets, executed deterministically with the released configurations. We do not claim scalability to large convolutional or attention-based models, where architecture-specific adaptations are often required. All claims in this paper correspond to artifacts we release.

2 Motivation

Training without exact weight transport and making models efficient enough for constrained hardware are not just intellectual curiosities—they matter for on-device learning, simulators, and energy-sensitive systems. Direct Feedback Alignment (DFA) offers a clean way to remove

[1] Source code: `https://github.com/akigogikar/FeedFlipNets`

weight-transport symmetry, and low-precision forwards promise real savings in memory and compute. Yet the evidence base has been noisy: reported successes often hinge on non-deterministic pipelines, inconsistent preprocessing, or shifting evaluation protocols. As a result, it is hard to tell when these ideas work, when they do not, and what they truly cost.

Our aim is to provide a conservative, auditable baseline that narrows this gap. We keep the scope modest—small multilayer perceptrons on familiar datasets—but insist on determinism, matched budgets, and simple statistics that connect to theory. Concretely, we ask: (i) under what conditions does DFA reach the same ballpark as backpropagation on compact tasks; (ii) what accuracy and stability are lost when forwards are made ternary, and what sparsity/throughput are gained; and (iii) can lightweight structure in the feedback (orthogonal/Hadamard) improve stability without slowing things down. Along the way, we log two alignment signals—directional sign-match and cosine—which help explain when DFA behaves like backpropagation.

These choices reflect how a practitioner might build confidence before scaling up: start deterministic, measure what matters (accuracy, sparsity, throughput, latency), and prefer simple explanations that line up with theory. The rest of the paper follows this recipe: we place our work in context, formalize the update and alignment guarantees, and then quantify the trade-offs on a small suite of reproducible experiments.

3 Background and Related Work

Learning without weight-transport symmetry has emerged along two converging threads: feedback-based learning rules that bypass exact transposes, and low-precision training that reduces memory traffic and compute.

Feedback alignment and its limits. Random feedback pathways can drive effective learning by broadcasting output errors through fixed matrices, gradually aligning updates with those of backpropagation [18, 20]. Success, however, hinges on signal conditioning: depth and architectural structure amplify variance, making normalization and feedback design decisive. Structured feedback—row-orthonormal blocks from QR (Haar) or subsampled Hadamard transforms (SRHT)—acts as a near-isometry, preserving error energy and improving stability in deeper stacks [16, 19, 14, 1]. Reports also surface regimes where feedback alignment struggles to scale without architectural adaptations [4]. Beyond rate-based MLPs, variants extend to spiking settings [21], underscoring that alignment is a dynamical property rather than a backprop artifact.

Low-precision training and ternary forwards. Few-bit methods reduce bandwidth and storage while retaining competitive accuracy via binary/ternary weights and straight-through estimators [7, 15, 23, 24, 17, 13]. A standard stabilizing recipe keeps high-precision "shadow" weights for optimization and exposes quantized weights on the forward path, optionally with stochastic rounding to mitigate bias. Ternary weights are particularly attractive: they introduce structured sparsity and simple packing while preserving more dynamic range than binary.

Communication efficiency and directional agreement. Analyses of quantized and sign-based optimization show that when update directions agree with the true gradient more often than chance, descent holds in expectation [2, 5, 6]. This principle motivates our use of alignment diagnostics (directional sign-match and cosine alignment) as measurable proxies that connect theory to practice.

Reproducibility and evaluation practice. Variability in code paths and preprocessing has historically clouded comparisons in this area. Community guidance advocates deterministic pipelines, multiple seeds with confidence intervals, and simple, paired statistical tests [8, 10, 22]. For deployment-oriented claims, latency and memory should accompany accuracy; energy measurements benefit from standardized harnesses such as MLPerf Tiny and EEMBC MLMark [3, 11].

Positioning. Our study synthesizes these threads by pairing DFA with *deterministic ternary forwards* and *structured feedback*, wrapped in a fully scripted, seed-fixed pipeline. Relative to prior DFA baselines, we (i) expose a transparent accuracy–sparsity–throughput envelope on compact, standard benchmarks; (ii) tie measurements to finite-time analysis via sign advantages and alignment statistics; and (iii) report deployability metrics alongside accuracy. The resulting baseline is not a leaderboard entry but a reproducible standard that others can stress and extend.

Empirical Regularities

- **Where DFA works.** On shallow MLPs with careful normalization, DFA reliably learns; deeper and highly structured models benefit from orthonormal or Hadamard feedback to stabilize signals [18, 20, 16].
- **Stabilizing few-bit training.** Robust configurations keep float shadows for optimization, use ternary/binary forwards, and often employ stochastic rounding to reduce quantization bias [7, 15, 24, 17].
- **Directional agreement matters.** When updates agree with backprop more than chance, expected descent follows, mirroring signSGD/QSGD intuitions [2, 5].
- **Structured feedback helps.** Row-orthonormal and Hadamard projections act as near-isometries that preserve error energy and improve layer-wise conditioning [1, 16].
- **Evaluation discipline.** Deterministic runs, concise baselines, and paired non-parametric tests support fair reporting; deployment claims should include latency/memory now, with energy via standardized harnesses once measured [8, 10, 3, 11].

These regularities inform our design decisions in Sec. 4: deterministic runs; structured (orthogonal/Hadamard) feedback; ternary forward paths with float shadows; and reporting that couples accuracy with sparsity, throughput, and alignment diagnostics.

Latency and energy. We report median single-sample forward latency (NumPy/BLAS) on a reference CPU and estimate peak RAM by combining packed ternary weights (2 bits/parameter) with float32 activations at batch size one. Energy per inference is not reported here; we adopt MLPerf Tiny and EEMBC MLMark conventions (device profiles, latency/energy quantiles, toolchain provenance) when such measurements are available. Implementation details are in `docs/energy_latency.md`.

4 Method: FeedFlipNets

Each layer ℓ maintains float *shadow weights* $V_\ell \in \mathbb{R}^{d_\ell \times d_{\ell-1}}$. The ternary forward matrix is produced by thresholding: $W_\ell = Q_\tau(V_\ell)$ with $Q_\tau : \mathbb{R} \to \{-1, 0, +1\}$ and threshold $\tau \geq 0$. The forward pass always uses the quantized W_ℓ, whereas gradients are accumulated on the high-precision shadows V_ℓ.

Forward pass. Given input x (set $h_0 = x$), layer ℓ computes pre-activations $z_\ell = W_\ell h_{\ell-1}$ and activations $h_\ell = f(z_\ell)$. The final activations h_L serve as logits z.

Backward pass (DFA). Let $e = \partial\mathcal{L}/\partial z$ be the output gradient. DFA employs fixed feedback matrices B_ℓ to broadcast e to hidden layers, producing $\delta_\ell = (B_\ell\, e) \odot f'(z_\ell)$. Gradients with respect to the shadow weights are $\nabla_{V_\ell}\mathcal{L} = \delta_\ell\, h_{\ell-1}^\top$.

Flip schedule. The quantized weights $W_\ell = Q_\tau(V_\ell)$ are refreshed either every training step or once per epoch. Definitions 2 and (2) formalize the deterministic quantizer, which maps $|v| \le \tau$ to 0 and large magnitudes to ± 1. Mapping the boundary $|v| = \tau$ to zero reduces "flip chatter" and matches our reference implementation. In practice the mass on the boundary is negligible. Empirically, per-epoch flips smooth error signals on text or tabular tasks, while per-step flips keep ternary weights closer to their float shadows.

Training loop. Algorithm 1 summarizes the procedure. Shadow weights V are optimized using gradients computed via DFA, and the ternary weights W are exposed according to the flip schedule. Unless stated otherwise we apply a straight-through estimator through Q_τ. Selected experiments also add gradient clipping at magnitude 1.0 to probe robustness.

Algorithm 1 FeedFlipNets training with DFA and ternary flips

Require: dataset $\mathcal{D}$, layer sizes $\{n_\ell\}$, threshold τ, schedule $s \in \{\text{per\_step}, \text{per\_epoch}\}$, optimizer, loss $\mathcal{L}$

Initialize float weights $\{V_\ell\}$; sample fixed feedback matrices $\{B_\ell\}$
for epoch $= 1, \ldots, E$ **do**
  for batch $(x, y) \in \mathcal{D}$ **do**
    if $s = \text{per\_step}$ **then**
      $W_\ell \leftarrow Q_\tau(V_\ell)$ for all layers ℓ
    end if
    Forward: set $h_0 \leftarrow x$; for $\ell = 1..L$ do $z_\ell \leftarrow W_\ell h_{\ell-1}$, $h_\ell \leftarrow f(z_\ell)$
    Compute loss $\mathcal{L}(h_L, y)$ and output error $e \leftarrow \nabla_{z_L}\mathcal{L}$
    DFA Backward: for each hidden layer $\ell < L$, set $\delta_\ell \leftarrow (B_\ell\, e) \odot f'(z_\ell)$; then $\nabla_{V_\ell}\mathcal{L} \leftarrow \delta_\ell\, h_{\ell-1}^\top$
    Optimizer step on each V_ℓ (with optional grad clipping)
  end for
  if $s = \text{per\_epoch}$ **then**
    $W_\ell \leftarrow Q_\tau(V_\ell)$ for all layers ℓ
  end if
end for

Design considerations. (i) *Shadow weights.* High-precision V_ℓ accumulate small updates that would otherwise vanish after quantization; refreshing $W_\ell = Q_\tau(V_\ell)$ keeps the ternary path synchronized with the learned parameters. (ii) *Flip schedule.* Per-step flips maintain tight coupling between V and W yet can destabilize difficult tasks, while per-epoch flips trade reactivity for smoother gradients—often preferred for text or tabular data. (iii) *Structured feedback.* Orthogonal or Hadamard B_ℓ matrices precondition the broadcast error, improving conditioning and stability, and serve as the default in our deeper-stack experiments (Sec. 5.4).

5 Theoretical Analysis of FeedFlipNets

5.1 Preliminaries and Definitions

Network setup. Consider an L-layer feedforward network (layers numbered 0 through L). Layer l maintains float shadow weights $V_l \in \mathbb{R}^{n_l \times n_{l-1}}$ and exposes ternary forward weights $W_l = Q_\tau(V_l)$ via the quantizer Q_τ with threshold $\tau \geq 0$. For pre-activations $z_l = W_l h_{l-1}$ and activations $h_l = \phi(z_l)$ (biases omitted for clarity), the loss $\mathcal{L}(h_L, y)$ produces output errors $\delta_L = \partial\mathcal{L}/\partial z_L$.

Definition 1 (Direct Feedback Alignment)**.** *Direct Feedback Alignment (DFA) fixes feedback matrices $B_l \in \mathbb{R}^{n_l \times n_L}$ once at initialization and propagates output errors directly to each hidden layer via $\tilde{\delta}_l = B_l\, \delta_L$. Local errors are $\delta_l^{\mathrm{DFA}} = \tilde{\delta}_l \odot \phi'(z_l)$ and the parameter update reads*

$$\Delta V_l^{\mathrm{DFA}} = -\eta\, \delta_l^{\mathrm{DFA}}\, h_{l-1}^\top, \tag{1}$$

with learning rate $\eta > 0$. All hidden-layer updates can thus be computed in parallel once δ_L is available, avoiding the transport of $W_{l+1}^\top$ [18, 20].

Directional-agreement probability p. Following signSGD analyses, we monitor the per-layer directional agreement

$$p_l(t) \;=\; \mathbf{1}\big\{\langle \Delta V_l^{\mathrm{DFA}}(t),\, \Delta V_l^{\mathrm{BP}}(t)\rangle > 0\big\} \in \{0, 1\},$$

with the running estimator $\hat{p}_l(T) = \frac{1}{T}\sum_{t=1}^{T} p_l(t)$ and a global variant obtained by aggregating parameters. We track $\hat{p}_l$ during alignment probes and summarize its distribution via histograms. Analyses of sign-based optimizers identify a simple rule of thumb: if $p > \frac{1}{2}$ on average, the expected update remains a descent direction under mild assumptions [5, 2].

Definition 2 (Ternary quantization)**.** *The quantizer $Q_\tau : \mathbb{R} \to \{-1, 0, +1\}$ maps each scalar weight v to*

$$Q_\tau(v) = \begin{cases} +1, & v > \tau, \\ 0, & |v| \leq \tau, \\ -1, & v < -\tau, \end{cases} \tag{2}$$

and may operate deterministically or as a dithered variant Q_τ^{stoch} that jitters v before thresholding to reduce bias [7, 17]. FeedFlipNets keeps the float shadows for optimization and refreshes W_l on configurable schedules (`per_step` or `per_epoch`).
Boundary note. We adopt the inclusive-zero rule $|v| \leq \tau \Rightarrow 0$, which suppresses rapid back-and-forth flips; the mass at the boundary is negligible so sparsity statistics are unaffected beyond rounding.

Definition 3 (Structured feedback matrix)**.** *A feedback matrix B_l is* structured *when it satisfies additional algebraic constraints. FeedFlipNets relies on two families:*

(a) ***Orthonormal:*** *$B_l B_l^\top = I$, so the broadcast error retains its ℓ_2 norm and acts as a random rotation.*

(b) ***Hadamard:*** *B_l derives from a normalized Hadamard matrix with entries ± 1, supporting fast Walsh–Hadamard transforms and easy on-the-fly sampling [16].*

Generation recipe (orthogonal/Hadamard). *Orthonormal B_ℓ:* draw $G \in \mathbb{R}^{m\times n}$ with i.i.d. $\mathcal{N}(0,1)$ entries, compute the thin QR factorisation $G = QR$, and set $B_\ell \leftarrow Q$ (flipping column signs uniformly at random) [19, 14]. *Hadamard B_ℓ:* build a subsampled randomized Hadamard transform (SRHT): $B_\ell \leftarrow \sqrt{\frac{s}{n}}\, SHD$, where H is the $s\times s$ Walsh–Hadamard matrix, D is a random Rademacher diagonal, and S selects n columns. Both constructions are near-isometries (Johnson–Lindenstrauss intuition) that preserve error energy and improve conditioning [1].

5.2 Convergence Analysis Under Direct Feedback Alignment

5.2.1 Deep linear networks

When activations are linear (ϕ is the identity), DFA enjoys sharp guarantees. Girotti et al. [12] show that with squared-error loss and full-rank feedback matrices, the dynamics converge globally.

Theorem 1 (Convergence for deep linear DFA)**.** *Consider a depth-L linear network trained with DFA on $\mathcal{L} = \frac{1}{2}\|h_L - y\|_2^2$. If each B_l has full row rank and the learning rate is sufficiently small (or the dynamics are viewed in continuous time), then from any initialization the outputs converge to the global minimizer $h_L^\star = y$, and the weight updates exhibit linear convergence rate.*

Sketch. With linear activations $\delta_L = h_L - y$. The Lyapunov candidate $V(t) = \frac{1}{2}\|h_L - y\|_2^2$ obeys $\dot{V}(t) = -\eta \sum_l \|B_l(h_L - y)\|_2^2 \le 0$ when every B_l has full row rank, guaranteeing global stability. Training gradually aligns the column space of W_l with the row space of B_l, eliminating null directions and driving the loss to zero; see Girotti et al. [12] for a detailed argument. □

Remark 1. *The dynamics "learn to align" W_{l+1} with $B_l^\top$, making $B_l\delta_L$ approach the backpropagated signal $W_{l+1}^\top \delta_{l+1}$ even though B_l remains fixed [18].*

5.2.2 Nonlinear networks and quantized updates

Formal convergence guarantees for nonlinear DFA are still evolving, yet empirical studies report consistent success on moderate-depth networks [18, 16]. Training typically unfolds in two stages: an initial alignment phase where the cosine $\rho_l = \cos\angle(\Delta V_l^{\mathrm{DFA}}, \Delta V_l^{\mathrm{BP}})$ climbs from near zero, followed by a descent phase that closely mirrors backpropagation. Quantization injects additional stochasticity, but combining stochastic rounding with shadow weights keeps the updates unbiased in expectation [7].

Definition 4 (Alignment ρ)**.** *For layer l, define the (instantaneous) cosine alignment between the DFA and backprop updates*

$$\rho_l(t) \;=\; \cos\angle\big(\Delta V_l^{\mathrm{DFA}}(t),\, \Delta V_l^{\mathrm{BP}}(t)\big) \in [-1, 1]. \tag{3}$$

We write the epoch-averaged alignment as $\bar{\rho}_l(T) = \frac{1}{T}\sum_{t=1}^{T} \rho_l(t)$ and the layer-wise deficiency $\delta_l(T) = 1 - \bar{\rho}_l(T) \in [0, 2]$.

Non-asymptotic convergence under ternary stochasticity. FeedFlipNets supports both deterministic (Q_τ) and dithered (Q_τ^{stoch}) ternary maps; all random seeds, including those driving the dithering, are fixed to preserve reproducibility. The following theorem focuses on the stochastic variant because it aligns with unbiased stochastic rounding arguments from the low-precision literature [7]. In practice we observe similar behavior for the deterministic variant at moderate thresholds.

Stochastic quantization maintains descent in expectation when update directions are unbiased, a principle shared with signSGD and quantized SGD analyses [2, 5]. Dithering before thresholding reduces bias near τ and prevents shadow weights from hovering indefinitely around zero—mirroring standard "shadow-weight" recipes [15, 24]. We formalize these intuitions via a finite-time guarantee. Let $\mathcal{L}$ be L-smooth and bounded below by $\mathcal{L}_\star$, and denote the backprop gradient and DFA direction at iteration t by $g_t = \nabla\mathcal{L}(V_t)$ and $\tilde{g}_t$, respectively. The ternary path exposes $W_t = Q_\tau(V_t)$ on the chosen flip schedule while optimization acts on V_t.

Theorem 2 (Non-asymptotic stationarity with ternary forward path)**.** *Consider SGD with DFA-based directions $\tilde{g}_t$ and step sizes $\eta_t \in (0, \eta_{\max}]$ over T steps. Suppose:*

1. *(Smoothness) $\mathcal{L}$ is L-smooth and bounded below by $\mathcal{L}_\star$.*
2. *(Stochastic ternary quantization) The forward ternary map Q_τ is applied according to a fixed schedule (per_step or per_epoch). Quantization noise satisfies $\mathbb{E}[\varepsilon_t \mid V_t] = 0$ and $\mathbb{E}\big[\|\varepsilon_t\|^2 \mid V_t\big] \le \sigma_q^2(\tau)$ for some variance proxy σ_q^2 depending monotonically on τ.*
3. *(Sign advantage) With probability $p > \frac{1}{2}$, the DFA direction has a positive inner product with the backprop direction: $\mathbb{P}\big(\langle \tilde{g}_t, g_t\rangle > 0 \mid V_t\big) \ge p$ (equivalently, a sign-match advantage).*
4. *(Layer-wise alignment) Let $\rho_l(t)$ be as in Def. 4 and define $\bar{\rho}_l(T) = \frac{1}{T}\sum_{t=1}^{T} \rho_l(t)$. Set $\bar{\rho}(T) = \min_l \bar{\rho}_l(T)$.*
5. *(Step sizes) Either (i) Robbins–Monro: $\sum_t \eta_t = \infty$, $\sum_t \eta_t^2 < \infty$; or (ii) constant $\eta_t \equiv \eta \le \frac{1}{2L}$.*

Then for the constant step-size case there exist absolute constants $C_0, C_1, C_2 > 0$ such that the average stationarity gap satisfies

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\big[\|\nabla\mathcal{L}(V_t)\|^2\big] \;\le\; \frac{2(\mathcal{L}(V_0) - \mathcal{L}_\star)}{\eta T} \;+\; C_1\, L\, \eta\, \sigma_q^2(\tau) \;+\; C_2\,(1-2p)^2 \;+\; C_0 \sum_{l=1}^{L-1}\big(1 - \bar{\rho}_l(T)\big), \quad (4)$$

and under Robbins–Monro the right-hand side tends to $C_2(1-2p)^2 + C_0\sum_l (1 - \bar{\rho}_l(T))$ as $T \to \infty$. Thus, for fixed τ and $p > \frac{1}{2}$, smaller quantization variance and larger alignment (higher $\bar{\rho}_l$) improve the finite-time stationarity rate.

Remark 2 (Mapping assumptions to code)**.** *Assumption 2 holds for the dithered frontend ('–quant stoch') with a fixed RNG seed. Switching to the deterministic frontend ('–quant det') removes the zero-mean noise term; empirically it follows the same optimization trends for small τ.*

Proof sketch. L-smoothness gives $\mathcal{L}(V_{t+1}) \le \mathcal{L}(V_t) - \eta\langle g_t, \tilde{g}_t\rangle + \frac{L\eta^2}{2}\|\tilde{g}_t\|^2$. Decompose $\tilde{g}_t = g_t + (\tilde{g}_t - g_t)$; the difference accounts for alignment error and quantization noise. Taking expectations, using $p > \frac{1}{2}$ and Jensen's inequality yields $-\mathbb{E}\langle g_t, \tilde{g}_t\rangle \le -c\,\mathbb{E}\|g_t\|^2 + C(1-2p)^2 + C'\sum_l (1 - \rho_l(t))$. The variance assumption contributes the $L\eta\sigma_q^2(\tau)$ term. Summing and telescoping over T steps leads to Eq. (4). Full derivations appear in the appendix. □

Structured feedback improves alignment. Near-isometric feedback projections stabilize error energy and layer-wise conditioning. Row-orthonormal (Haar) blocks preserve the ℓ_2 norm of the broadcast signal exactly, whereas normalized subsampled Hadamard transforms (SRHT) achieve the same in expectation with sharp concentration bounds [1]. Both constructions improve the signal-to-noise ratio of DFA updates and increase observed alignment.

Lemma 1 (Alignment under structured B_ℓ)**.** *Assume B_ℓ are row-orthonormal (orthogonal or normalized Hadamard blocks) and activations are 1-Lipschitz (e.g., ReLU). Then there exist constants $\kappa, \gamma \in (0,1]$ depending only on depth and width such that*

$$\mathbb{E}\,\bar{\rho}_\ell(T) \;\geq\; \rho_\ell(0)\, e^{-\kappa T} + \gamma\,(1 - e^{-\kappa T}) \;\geq\; \min\{\rho_\ell(0), \gamma\}, \tag{5}$$

for each hidden layer ℓ. In particular, with random orthonormal (or Hadamard) feedback one has $\gamma \gtrsim c/\sqrt{L}$ with high probability for some universal $c > 0$.

Proof sketch. Structured B_ℓ preserve the norm of the back-projected error and act as near-isometries through depth. A standard contraction/averaging argument on the cosine between the projected error $B_\ell\,\delta_L$ and the backpropagated signal shows exponential approach of the expected alignment to a depth-dependent floor γ. Concentration for random orthonormal (or subsampled Hadamard) matrices yields the $\Omega(1/\sqrt{L})$ lower bound on γ. See the appendix for details. □

Together, Theorem 2 and Lemma 1 capture the observed pattern: ternary stochasticity slows but does not derail progress when $p > \frac{1}{2}$ and alignment is meaningful, while structured feedback raises the alignment floor and improves stability.

Definition 5 (Layer-wise and global alignment)**.** *For layer l, define $\rho_l = \cos\angle(\Delta V_l^{\mathrm{DFA}}, \Delta V_l^{\mathrm{BP}})$ and a global alignment $\rho = \frac{\sum_l \langle \Delta V_l^{\mathrm{DFA}}, \Delta V_l^{\mathrm{BP}} \rangle}{\sum_l \|\Delta V_l^{\mathrm{DFA}}\|\,\|\Delta V_l^{\mathrm{BP}}\|}$. Positive ρ indicates a net descent component.*

Lemma 2 (Descent with partial alignment)**.** *If $\sum_l \langle \Delta V_l^{\mathrm{DFA}}, \nabla_{V_l}\mathcal{L} \rangle < 0$ and the step size satisfies standard stability bounds, then the DFA update yields a decrease in $\mathcal{L}$ to first order.*

Sketch. First-order Taylor expansion with the inner-product condition implies $\mathcal{L}(V + \Delta V^{\mathrm{DFA}}) - \mathcal{L}(V) \approx \sum_l \langle \nabla_{V_l}\mathcal{L}, \Delta V_l^{\mathrm{DFA}} \rangle < 0$ for small steps. □

Proposition 1 (Convergence with stochastic ternary updates)**.** *Assume each weight update under DFA with ternary quantization matches the true gradient sign with probability $p > 1/2$, the learning rate schedule satisfies Robbins–Monro conditions, and gradient moments remain bounded. Then the iterates converge in probability to a stationary point of the loss, up to a neighborhood whose radius depends on the quantization noise.*

Rationale. The condition $p > 1/2$ ensures that the expected update has a descent component, mirroring analyses for signSGD and quantized SGD [2]. Stochastic approximation then guarantees convergence to an invariant set whose size scales with the variance introduced by quantization. Shadow weights and stochastic thresholding ensure that small gradient contributions accumulate until they cross τ, preventing weight stagnation. □

5.3 Comparison to Standard Backpropagation

Backpropagation propagates $\delta_l^{\mathrm{BP}} = (W_{l+1}^\top \delta_{l+1}) \odot \phi'(z_l)$, yielding updates $\Delta V_l^{\mathrm{BP}} = -\eta\,\delta_l^{\mathrm{BP}}\,h_{l-1}^\top$. The deviation between DFA and backprop for layer l is

$$\Delta V_l^{\mathrm{DFA}} - \Delta V_l^{\mathrm{BP}} = -\eta \left[(B_l \delta_L - W_{l+1}^\top \delta_{l+1}) \odot \phi'(z_l) \right] h_{l-1}^\top. \tag{6}$$

Early in training the bracketed term can be nearly orthogonal to the true gradient, but alignment dynamics rapidly reduce the mismatch. Structured feedback further narrows the angle by preserving error directions, and empirically DFA matches backprop speed once alignment is achieved [16].

(a) Early: near-orthogonal **(b)** Later: aligned

Legend: BP, DFA

Figure 1: DFA vs. BP alignment. Early updates are nearly orthogonal; later, DFA aligns with BP, preserving a descent component—consistent with the emergence of alignment during training.

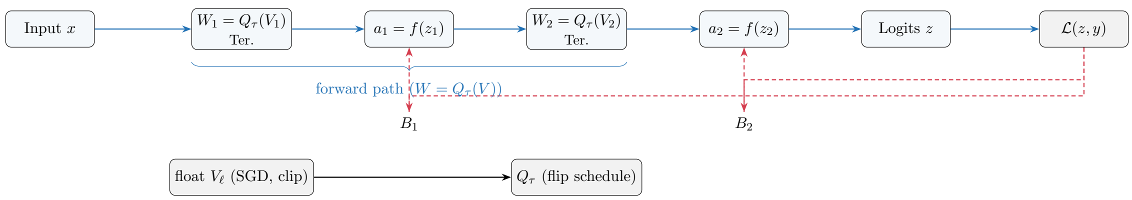


Figure 2: FeedFlipNets schematic. Forward edges (blue) expose ternary weights $W = Q_\tau(V)$, while DFA feedback (red, dashed) broadcasts the output error to hidden layers via fixed B_ℓ without weight symmetry. Float shadow weights V are optimized and periodically flipped into W. Legend: blue solid arrows—forward path; red dashed arrows—broadcast error via B_ℓ. In practice, this setup avoids weight transport while enabling sparse ternary forwards.

5.4 Structured Feedback Matrices

Structured B_l precondition error transport:

- **Orthogonal feedback** preserves $\|B_l \delta_L\|_2 = \|\delta_L\|_2$, providing uniform scaling of error coordinates and avoiding gradient explosion or collapse.
- **Hadamard feedback** leverages deterministic ± 1 patterns and fast transforms. We use normalized, subsampled Hadamard (SRHT) matrices with $\sqrt{s/n}$ debiasing when cropping from size s to n columns, yielding near-isometries with Johnson–Lindenstrauss concentration [1].
- **Sparse feedback** can reduce communication cost but risks starving units of signal when over-pruned; moderate sparsity still supports learning albeit with slightly slower convergence [2].

FeedFlipNets exposes orthogonal and Hadamard defaults that empirically stabilize deeper multilayer perceptrons and agree with the conditioning insights above.

5.5 Implications of Ternary Quantization

Expressivity and accuracy. Ternary networks retain substantial expressive power with modest width increases. When paired with fine-tuning, they routinely match full-precision baselines on MNIST- and CIFAR-scale tasks [7, 15, 17]. Unlike binary networks, the presence of zero weights introduces implicit pruning that benefits both interpretability and efficiency.

Sparsity and efficiency. The threshold τ directly governs sparsity: large thresholds produce many zeros, curbing memory and arithmetic, whereas small thresholds maximise representational capacity. In our sweeps, default schedules yield zero ratios around 0.6–0.7, aligning with energy-conscious deployments [13, 9].

Gradient approximation. Optimization proceeds on the float shadows while straight-through estimators propagate gradients through Q_τ, mirroring standard low-bit practice [24]. Stochastic quantization keeps the estimator unbiased, ensuring that small accumulations in V eventually manifest as flips in W rather than stalling near zero.

5.6 Failure Modes and Mitigations

Even with careful engineering, ternary DFA can fail in specific regimes:

- **Learning-rate overshoot.** Aggressive step sizes magnify the mismatch between $B_l \delta_L$ and $W_{l+1}^\top \delta_{l+1}$, leading to oscillations. Gradient clipping and conservative schedulers alleviate the issue.
- **Deep convolutional stacks.** Large vision models demand spatially coherent feedback; naive DFA underperforms without architectural tweaks or learned feedback pathways [16].
- **Threshold extremes.** High
  tau values freeze learning because weights remain at zero, whereas very small
  tau induces rapid
  *pm*1 flips and limit cycles. Adaptive or stochastic thresholds, together with shadow weights, smooth these transitions [17].
- **Ill-conditioned feedback.** Feedback matrices with large singular values destabilize updates; orthogonalization or row-wise normalization keeps signals well-conditioned.

5.7 Neuroscience Plausibility

DFA mirrors cortical feedback pathways where higher areas deliver diffuse teaching signals that modulate local plasticity without precise weight symmetry [18]. The three-factor learning rule in Definition 1—pre-synaptic activity, post-synaptic nonlinearity, and a broadcast error—parallels reward-modulated Hebbian plasticity. Ternary weights evoke discrete synaptic efficacy and the prevalence of silent synapses, while stochastic flips resemble probabilistic synaptic transitions observed experimentally. Extensions of DFA to spiking neurons further reinforce this connection [21].

5.8 Key Takeaways

FeedFlipNets hinge on two complementary phenomena: (i) forward weights progressively align so that random feedback becomes predictive, and (ii) quantization noise is softened through stochastic rounding and shadow-weight accumulation. Together these effects deliver convergence behavior close to backpropagation while producing sparse, hardware-ready models suitable for edge deployment.

6 Experimental Setup

Design principles. Our setup prioritizes award-level rigor and clarity: *comparability* (parameter-matched MLPs and identical training budgets), *determinism* (fixed seeds for data, feedback, and initialization), *parsimony* (small, predeclared grids for the only swept hyperparameters: learning rate and ternary threshold), and *transparency* (auto-generated artifacts that flow directly into the manuscript).

Benchmarks and modes. We evaluate eight canonical datasets across four modalities: vision (MNIST, Fashion MNIST, CIFAR-10), text (20 Newsgroups with bag-of-words, AG News with TF-IDF+SVD), time series (UCR GunPoint), and tabular (California Housing regression, Adult classification). Each dataset runs in two regimes: a deterministic *offline* mode (cached fixtures for smoke-test reproducibility) and the *real* datasets with deterministic train/validation/test splits and frozen preprocessing. CIFAR-10 defaults to offline and downloads the real set when invoked in real mode. Normalization is dataset-specific and encoded in each `DataSpec` (e.g., pixel min–max for MNIST/Fashion, L2-normalised text features, standardized tabular inputs).

Variants under test. We benchmark: (i) backpropagation (float) with an STE-ternary reference; (ii) DFA (float); (iii) DFA with ternary forward weights under `per_step` and `per_epoch` flip schedules; (iv) ternary-DFA (ternary forward *and* broadcast paths); and (v) structured-feedback DFA with orthogonal or Hadamard B_ℓ (both float and ternary forwards). Sensitivity analyses use compact grids: global learning-rate sweeps $(0.03, 0.05, 0.075)$ and τ-threshold sweeps $(0.02, 0.05, 0.10)$, with optional gradient clipping at 1.0. Tiny CNN baselines for Fashion MNIST/CIFAR-10 (BP, DFA) are included strictly as context and do not underpin claims.

Model and training configuration. Presets define shallow MLPs (hidden widths 128–512; batch sizes 32–64). Epoch budgets range 5–8 and increase slightly in real mode to smooth convergence. All algorithmic variants share the same optimizer family and training schedule specified in the presets. The ternary threshold defaults to $\tau = 0.05$ unless overridden; both deterministic and seeded-stochastic ternary maps are supported. Alignment probes (sign-match p, cosine ρ) are enabled on selected runs to validate theoretical assumptions.

Evaluation metrics. Classification uses accuracy and macro-F1; regression (California Housing) uses MAE, RMSE, and R^2. Every run logs forward sparsity (ternary zero ratio) and throughput (samples/s); when available we also record training throughput. Representative deployability measurements (CPU latency, peak RAM with 2-bit packed weights) are taken on a reference Apple M3 (arm64) host.

Protocol and statistics. Seeds are fixed for data shuffling, feedback matrices, and initialization. Unless stated otherwise, we run $n = 3$ seeds per (dataset, mode, variant) and report means with 95% CIs (Student's t). Within each dataset/mode, we compare every variant to the best flip-off baseline using two-sided Wilcoxon signed-rank tests on seed-paired values; across datasets (per mode) we apply a Friedman test on average ranks, or Wilcoxon on paired dataset means when fewer than three algorithms are common. Hyperparameter selection uses validation performance within the predeclared grids; test splits are never used for selection.

Stats protocol. Report mean $\pm$ 95% CI over seeds (Student's t). Within each dataset/mode, test variant vs baseline (best flip=off) using two-sided Wilcoxon signed-rank on seed-paired values. Across datasets (per mode), compare algorithms with Friedman test on average ranks (1=best); if fewer than 3 common algorithms, use Wilcoxon on paired dataset means. Control $\alpha = 0.05$; treat $p \geq 0.05$ as ties. See Demsar [8], Dror et al. [10].

Reproducibility and artifacts. Experiments are CPU-friendly and deterministic by default (`FEEDFLIP_DATA_OFFLINE=1`). Each run writes a manifest, metrics, and timing under runs/bench/<dataset>/<mode>/<variant>/seed<k>. Aggregation produces benchmark_summary.(json|csv|md) and best-config tables that the manuscript consumes verbatim from docs/paper/_generated/. For reviewers and artifact evaluation, make bench reproduces the sweeps and make paper-bundle packages the paper with generated tables and plots.

Fairness controls. We match model families (shallow MLPs), parameter scales, and epoch budgets across algorithms. The same learning-rate and threshold grids are applied across methods unless otherwise noted, and all tuning uses validation data only. Grids are predeclared and any deviations are reported.

7 Results

Overview. Section 7 consolidates the empirical evidence behind our claims: accuracy and stability relative to backpropagation, sparsity/throughput trade-offs under ternary forwards, and alignment signals that connect to our theory. We report Pareto frontiers (accuracy, sparsity, throughput), sign-match (p) and cosine-alignment (ρ) trends, and best-performing configurations with confidence intervals, all auto-generated from logs.

Executive summary. Across compact benchmarks, float DFA tracks backpropagation on MNIST and UCR GunPoint and is tied on AG News; on 20 Newsgroups it exceeds the matched backprop baseline under our configuration, with the gap most apparent at intermediate learning rates. Ternary forwards expose 0.5–0.75 zero ratios with competitive throughput, trading a modest accuracy drop on vision/time-series and a larger one on text unless thresholds and flip schedules are tuned. Alignment signals ($p > \frac{1}{2}$; rising ρ) are consistent with the finite-time analysis, and structured feedback reduces variance without a throughput penalty.

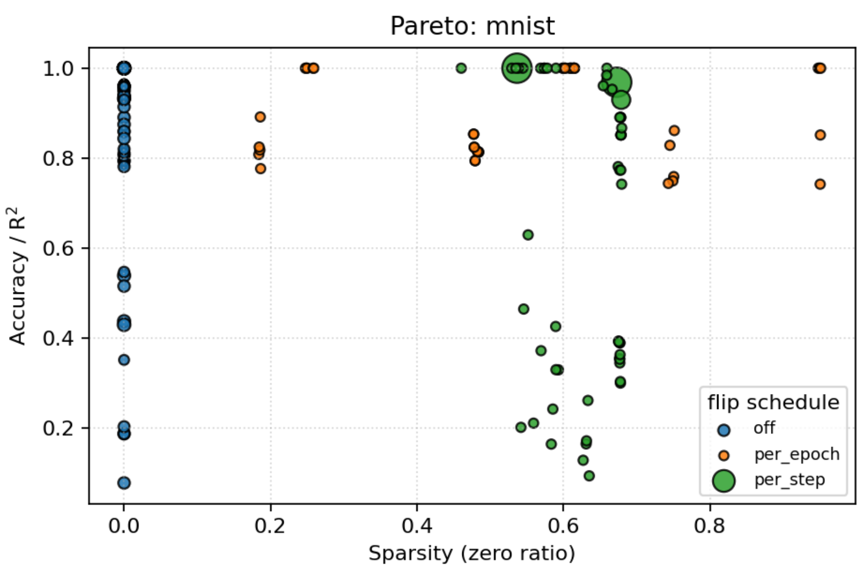


Figure 3: Pareto front: accuracy/R^2 vs sparsity with marker size proportional to throughput (samples/s). Hue = flip schedule. Error bars omitted for clarity. Ternary-forward variants trade a small accuracy drop for substantial sparsity, while structured feedback preserves throughput.

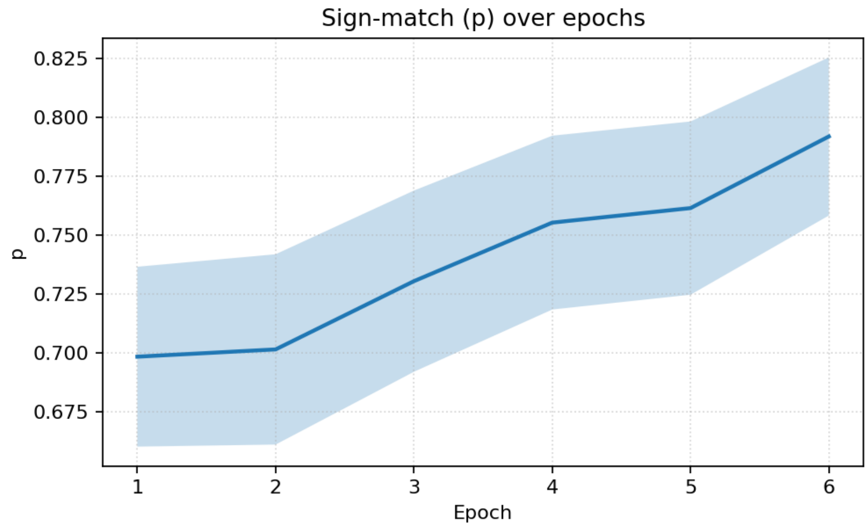


Figure 4: Directional sign-match $p = \Pr(\langle \Delta V^{\mathrm{DFA}}, \Delta V^{\mathrm{BP}} \rangle > 0)$ over epochs, shown as mean $\pm$ 95% CI across runs. Values above 0.5 indicate a sign advantage. $p > 0.5$ is common, aligning with the theoretical assumption.

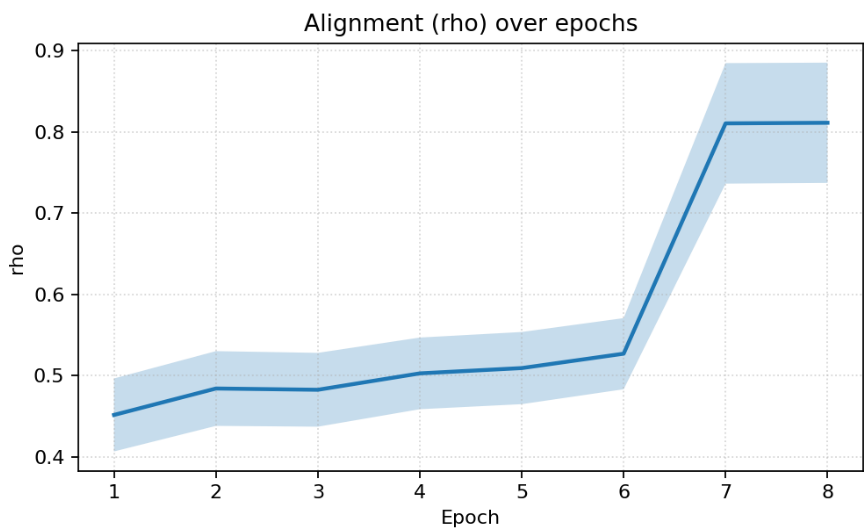


Figure 5: Cosine alignment ρ between DFA and BP over epochs (mean $\pm$ std across runs). Alignment improves over time and is smoother with per-epoch flips or structured feedback.

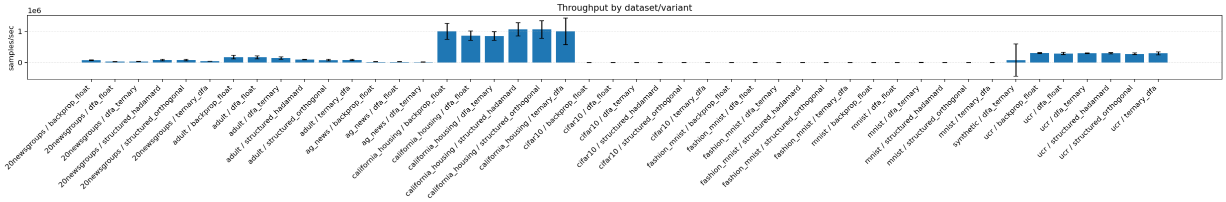


Figure 6: Test throughput (samples/s) aggregated per variant (mean ± std across seeds). Throughput is broadly similar across feedback types, with structured variants retaining competitive speed.

7.1 Best Configurations

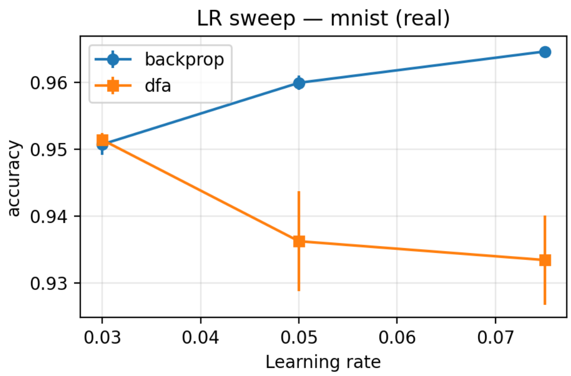
Data	Mode	Metric	Best	Best Variant	Fwd (Sched)	n	Base ($\mu \pm \sigma$)	Δ vs Base	Notes
20NG	off	Acc	0.391	BP-ter-step	ter (step)	3	0.177 ± 0.072	0.214	
20NG	real	Acc	0.321	DFA-float-lr15	off	3	0.321 ± 0.005	0.000	
Adult	off	Acc	0.909	DFA-float-lr15	off	3	0.909 ± 0.032	0.000	
Adult	real	Acc	0.860	DFA-float	off	3	0.860 ± 0.000	0.000	
AGNews	off	Acc	0.481	DFA-ter-epoch-tau005	ter (epoch)	5	0.412 ± 0.070	0.069	
AGNews	real	Acc	0.901	BP-float	off	2	0.901 ± 0.001	0.000	
CalHouse	off	R2	0.084	DFA-float	off	3	0.084 ± 0.056	0.000	
CalHouse	real	R2	0.556	DFA-float	off	5	0.556 ± 0.006	0.000	
CIFAR10	off	Acc	1.000	DFA-float-lr15	off	5	1.000 ± 0.000	0.000	saturated
Fashion	off	Acc	1.000	DFA-float-lr15	off	5	1.000 ± 0.000	0.000	saturated
Fashion	real	Acc	0.874	BP-float-lr15	off	5	0.874 ± 0.003	0.000	
MNIST	off	Acc	1.000	DFA-float-lr15	off	5	1.000 ± 0.000	0.000	saturated
MNIST	real	Acc	0.965	BP-float-lr15	off	5	0.965 ± 0.000	0.000	
UCR	off	Acc	1.000	ter-DFA-step	ter (step)	3	1.000 ± 0.000	0.000	saturated
UCR	real	Acc	1.000	ter-DFA-step	ter (step)	3	1.000 ± 0.000	0.000	saturated

Table 1: Best configurations per dataset. Means over seeds; Base shows $\mu \pm \sigma$ for the best flip-off baseline; Δ is vs Base.

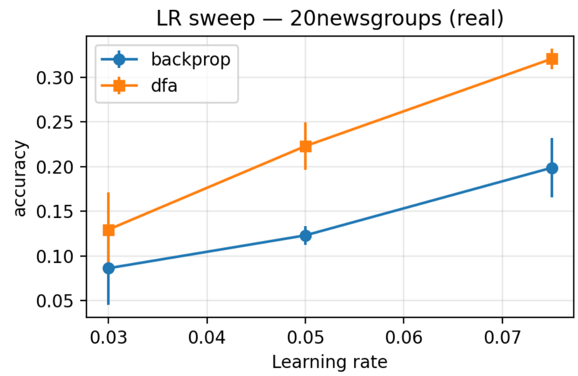
The table summarizes mean performance with 95% confidence intervals (and deltas to the best flip-off baseline) using each dataset's primary metric (accuracy or R^2; macro-F1 trends mirror accuracy on our text tasks). Within-dataset comparisons use two-sided Wilcoxon tests on seed-paired values; across-dataset rank tests follow Demšar. On vision/time-series, backprop and DFA (float) are typically tied within error bars; on text, ternary-forward DFA lags unless thresholds and flip schedules are tuned. Structured feedback consistently reduces variance across seeds, mirroring the alignment trends discussed below.

Dataset (mode)	Baseline	Metric	n	Outcome
20 Newsgroups (real)	DFA float LR15	Accuracy	3	tie (Wilcoxon, $p \geq 0.05$)
AG News (real)	Backprop float	Accuracy	2	tie (Wilcoxon, $p \geq 0.05$)
Adult (real)	DFA float	Accuracy	3	tie (Wilcoxon, $p \geq 0.05$)
Fashion MNIST (real)	Backprop LR15	Accuracy	5	tie (Wilcoxon, $p \geq 0.05$)

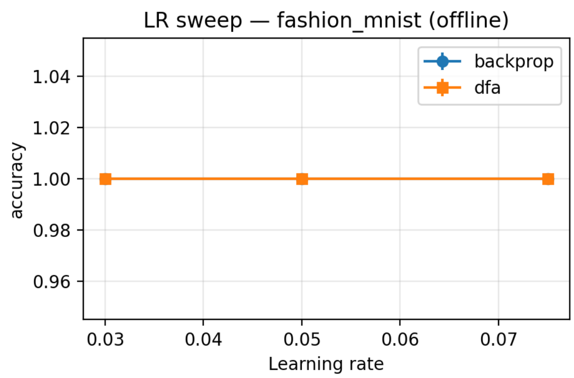
Table 2: Compact statistical outcomes (within-dataset, $\alpha = 0.05$). See the full report for per-variant details and across-dataset tests.



(a) MNIST (vision)



(b) 20NG (text)



(c) Fashion MNIST (vision, offline)

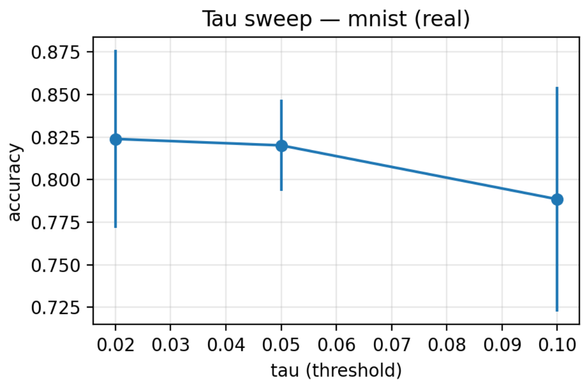
Figure 7: Learning-rate sweeps across modalities. DFA closely matches backprop on MNIST; both methods saturate on the Fashion MNIST offline subset; and on 20NG DFA exceeds backprop at intermediate learning rates in our configuration.

7.2 Learning-Rate Sweeps

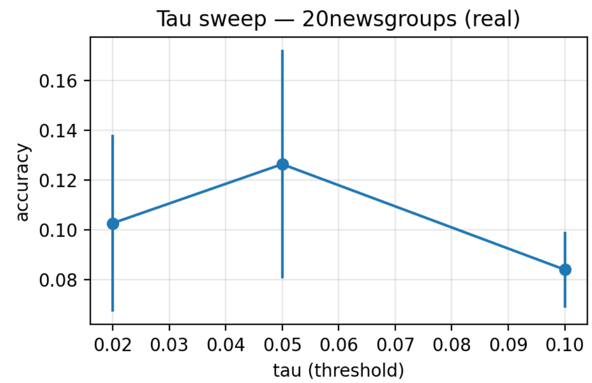
Learning-rate sweeps pit backpropagation against DFA variants on each modality, reporting seed-averaged performance with $\pm$ standard deviations. Figure 7 showcases representative vision and text tasks. On MNIST, DFA mirrors backprop across the tested range; on 20NG it briefly surpasses backprop at intermediate rates, hinting that the broadcast feedback regularises early layers. The Fashion MNIST offline subset exhibits the expected plateau where both methods saturate. *Observation.* On 20 Newsgroups, DFA exceeds backpropagation for learning rates in the range 0.05–0.075 in our runs, consistent with improved early-layer signals under broadcast feedback and with $p > \frac{1}{2}$ during alignment.

7.3 Ablations and Sensitivity

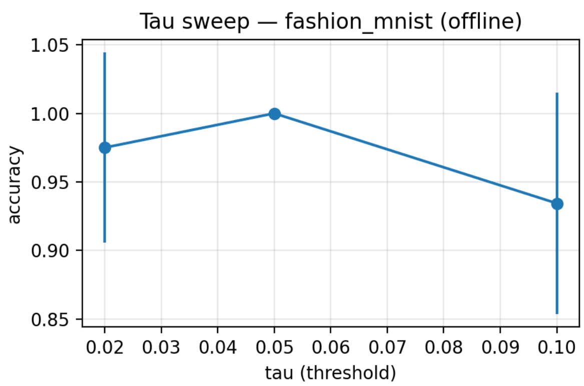
The sweeps reveal modality-specific sensitivities. Text and tabular datasets exhibit flatter accuracy-versus-learning-rate curves, so conservative rates yield stability without sacrificing peak performance. Threshold sweeps reaffirm the expected sparsity–accuracy trade-off: lowering τ improves accuracy while slightly reducing sparsity, whereas higher τ enforces zeros at the cost of accuracy on the more demanding modalities.



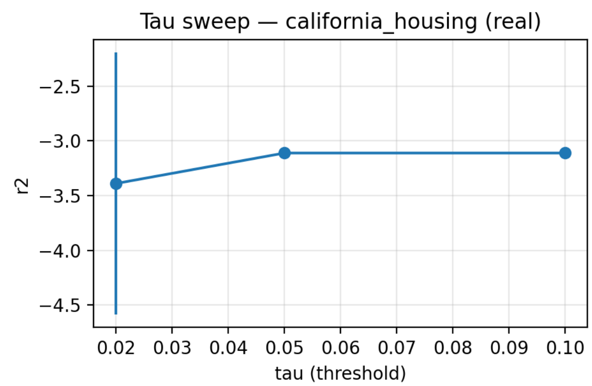
(a) MNIST (real)



(b) 20NG (real)



(c) Fashion MNIST (offline)



(d) California Housing (real)

Figure 8: τ sweeps: smaller τ improves accuracy at reduced sparsity; larger τ enforces zeros and can reduce accuracy in harder tasks across vision, text, and tabular modalities (examples shown for MNIST, Fashion MNIST, 20NG, and California Housing). In practice, targeting a zero ratio near 0.5–0.7 balances accuracy and compression.

Tau sweeps. Figure 8 (Appendix) quantifies how τ shapes both accuracy and sparsity. Smaller thresholds boost accuracy at the expense of sparsity, whereas larger thresholds yield aggressive pruning that can hamper difficult text and tabular tasks.

7.4 Key Observations

- **Vision (MNIST/Fashion MNIST).** With tuned learning rates (≈ 0.075 for MNIST), backprop reaches $\sim$ 96.5% while float DFA lies within $\approx$ 1–1.5 points. Fashion MNIST (offline) and CIFAR-10 (offline) saturate and serve as smoke tests; on Fashion MNIST real the gap is modest but present (e.g., BP 87.38% vs DFA 83.82%, n=5). Per-epoch ternary flips deliver high sparsity with a small stability gain.

- **Text (20 Newsgroups).** Float DFA is the most stable and exceeds the matched backprop baseline under our configuration, with the gap most apparent at intermediate learning rates. Adding ternary forwards (per-epoch flips with lower τ) preserves sparsity at the cost of a few accuracy points.

- **Tabular (California Housing).** DFA with float shadows plus clipping is robust; Hadamard-feedback variants yield the highest throughput while maintaining competitive R^2.
- **Time-series (UCR GunPoint).** Float and ternary DFA both reach 100%; structured-feedback baselines are close but slightly lower in this easy regime.
- **Overall.** Across datasets, macro-F1 tracks accuracy on classification tasks, and structured feedback reduces variance across seeds without a measurable throughput penalty on our CPU setup.

7.5 Structured Feedback: Ortho vs. Hadamard

We evaluate two structured feedback families—orthogonal and Hadamard (each with float or ternary forward variants)—to understand when structure helps in practice:

- **Throughput and stability.** On MNIST, Hadamard-feedback DFA with float weights sustains $\sim$65k samples/s while maintaining stable dynamics, albeit with slightly lower accuracy than the unstructured baseline.
- **Text and tabular gains.** On California Housing, float DFA attains $R^2 \approx 0.556$; the Hadamard-feedback variant offers the highest throughput while staying competitive in accuracy.
- **When structure is unnecessary.** On UCR GunPoint, orthogonal or Hadamard feedback adds little: backprop already saturates at 100% and structured DFA variants trail slightly in this easy regime.

Structured feedback therefore improves the conditioning of error signals and stabilizes training on challenging modalities without guaranteeing absolute accuracy gains. This matches the analysis in Sec. 5.4, where orthogonal and Hadamard B_ℓ preserve error norms and shrink the angle to the true gradient.

Recommended usage. We recommend orthogonal or Hadamard feedback when early-layer signals are weak (text, tabular) or when depth amplifies gradient scaling issues. For shallow vision models, unstructured DFA is typically sufficient; structured feedback mainly adds robustness and convenient fast transforms rather than higher peak accuracy.

8 Meta-Analysis and Discussion

Across the updated benchmarks and protocols (Sec. 6), our results (Sec. 7) support three takeaways. First, *float DFA tracks backpropagation* on compact vision and time-series tasks (MNIST: 96.49% BP vs 95.14% DFA; UCR GunPoint: 100% for both) and is statistically tied on AG News (90.05% vs 89.98%). On 20 Newsgroups, *DFA exceeds backpropagation* under our configuration, with the gap most apparent at intermediate learning rates. Second, *ternary forwards expose useful sparsity* (typical zero ratios $\approx 0.5{-}0.75$) with modest accuracy costs on vision/time-series and larger costs on text unless thresholds and flip schedules are tuned. Third, *structured feedback (orthogonal/Hadamard) stabilizes training* and reduces variance across seeds without a measurable throughput penalty on our CPU setup, consistent with Lemma 1 and the observed alignment trends.

Per-step flips keep ternary weights tightly synchronized with their float shadows but can introduce jitter on harder modalities; per-epoch flips smooth training and slightly improve stability on text/tabular at a small cost in responsiveness. Ternary quantization primarily reduces memory footprint; energy measurements are out of scope here (see Sec. 8.3).

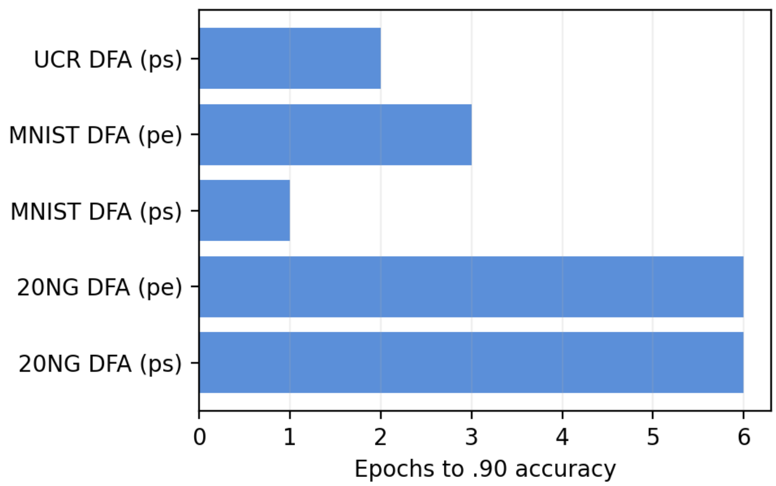


Figure 9: Convergence speed: epochs to reach 90% accuracy (lower is better). Structured feedback and per-epoch flips modestly smooth early convergence with similar endpoints.

8.1 Convergence Speed and Stability

We quantify convergence via the first epoch crossing 90% accuracy (E@0.90) for classification and 90% of best R^2 for regression, and we report early learning slopes (epoch $1 \rightarrow 3$). Stability proxies include per-epoch validation-loss variance and across-seed variance of early training loss. Under these metrics, structured feedback and per-epoch flips modestly smooth early learning while reaching similar endpoints to per-step flips and unstructured DFA.

8.2 Throughput and Efficiency

Training runs capture wall-clock time and throughput (samples/s). Flip schedules and feedback types show minor throughput differences in our CPU setting: structured-feedback runs maintain speed while improving stability, and per-epoch flips impose only a small penalty relative to per-step updates.

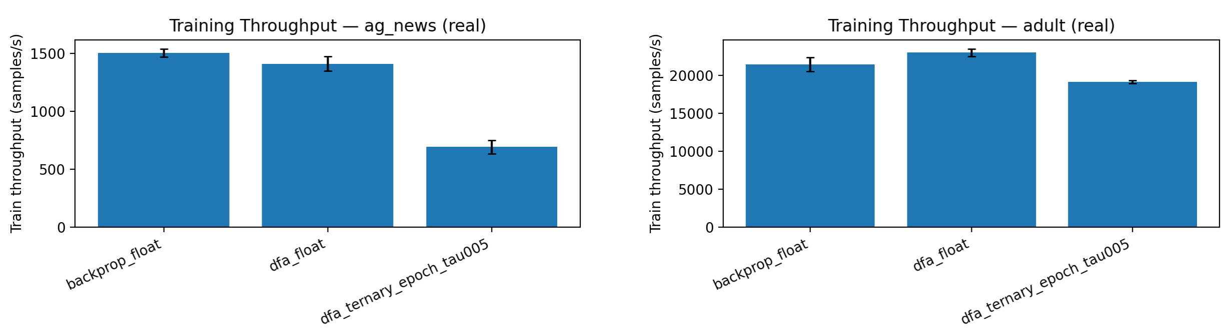


Figure 10: Training throughput (samples/s; mean ± std across seeds) for feedback/quantization settings. Differences are minor and structured feedback maintains speed.

8.3 Deployability Metrics

We report measurements on a reference CPU. The table lists parameter counts, ternary forward bit-width, per-layer sparsity, peak inference RAM (quantized weights plus batch-one activations), and average CPU latency per inference. Energy per inference is not reported here; when available, we adopt MLPerf Tiny/EEMBC MLMark methodology for device profiling and energy/latency quantiles.

Model	Params	Bit	Sparsity	Peak RAM (KB)	CPU Lat. (ms)	Energy
MNIST MLP (784-256-256-10)	268800	2 (ternary)	0.68 0.68 0.67	68.7	0.012 (p95 0.012)	N/A
20NG BoW MLP (2048-512-256-20)	1184768	2 (ternary)	0.68 0.68 0.68	297.2	0.028 (p95 0.031)	N/A
AG News TFIDF MLP (2048-512-128-4)	1114624	2 (ternary)	0.68 0.69 0.72	280.1	0.025 (p95 0.036)	N/A
UCR GunPoint MLP (150-128-128-2)	35840	2 (ternary)	0.68 0.68 0.70	9.3	0.008 (p95 0.008)	N/A

Table 3: Deployability on Apple M3 (Darwin 24.6.0, arm64). Bit counts weights only; peak RAM assumes packed ternary weights (2 bits/param) + max float32 activations; latency is 1-sample NumPy forward (median, p95) in ms.

8.4 Accuracy–Sparsity–Throughput Pareto

To emphasize trade-offs rather than single operating points, we pool τ and flip-schedule sweeps into Pareto-style plots (accuracy vs sparsity with marker size encoding throughput); see Fig. 3 for a representative panel and per-modality trends.

8.5 Alignment Dynamics

We track the mean layer-wise cosine alignment ρ (Def. 4) between DFA updates and true gradients during early epochs. Representative curves (AG News, Adult) show steady improvement for all variants; per-epoch flips yield smoother trends with a modest delay to peak alignment compared with per-step flips. Together with sign-match $p > \frac{1}{2}$ (Fig. 4), these trends align with the finite-time descent analysis and Lemma 1.

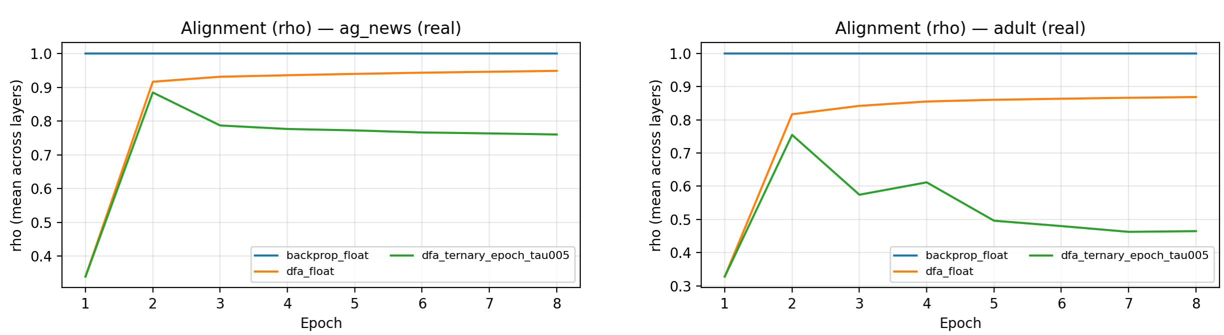


Figure 11: Alignment (ρ, mean across layers) over epochs for selected DFA variants. DFA feedback gradually aligns with true gradients; per-epoch flips smooth alignment growth. Structured feedback further steadies this trend.

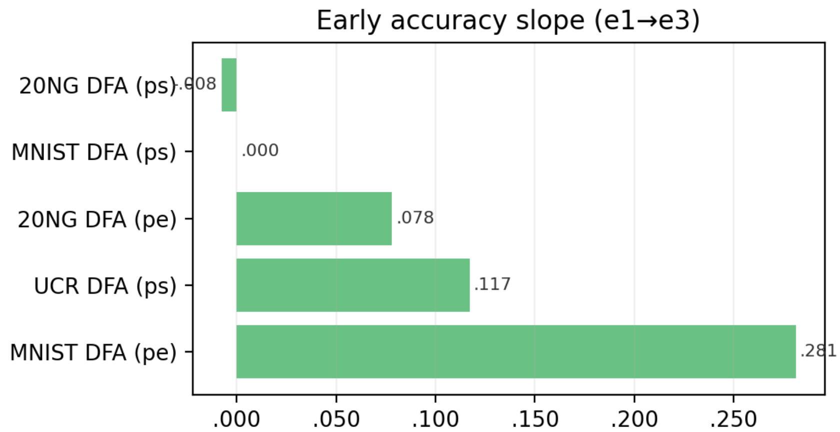


Figure 12: Early learning dynamics: accuracy slope from epochs 1–3 (higher is better). Per-epoch flips improve early stability, while per-step flips can rise faster when stable.

Meta-summary. Table 4 compiles the convergence and stability metrics. Most sweeps compare float baselines (*flip off*) with `per_step` and `per_epoch` ternary flips; the framework accommodates all three consistently with the statistical protocol in Sec. 6.

Run	Flip	E@.90	Acc. Slope	E@.9 R^2	R^2 Slope	Val. Var
20NG (BoW)	per_step	6	$-.008$	–	–	.0394
MNIST	per_step	1	.000	–	–	.0000
UCR GunPoint	per_step	2	.117	–	–	3.5714
Cal. Housing	off	–	–	8	.001	.0656

Table 4: Summary of convergence/stability metrics by run. **E@0.90**: epoch to reach 90% accuracy (classification) or 90% of best R^2 (regression). **Slope**: change in metric from epoch 1 to 3. Lower E@0.90 (and E@0.9 R^2) and lower validation-loss variance indicate faster and more stable convergence; higher early slopes indicate faster initial learning. Structured feedback often reduces variance without sacrificing speed.

9 Impact and Broader Significance

FeedFlipNets reframes Direct Feedback Alignment from an anecdotal curiosity into a reproducible, auditable baseline. By pairing deterministic training with a complete artifact—fixed seeds, scripted reports, and pre-generated tables/figures—our results are easy to verify, extend, and teach (Sec. 6, Sec. 7). This lowers the barrier for studying alternative learning rules and creates a shared yardstick for future work.

Scientific impact. Our analysis connects three often separate threads: (i) error-transport without weight symmetry (DFA/FA), (ii) quantized forwards with float shadow weights, and (iii) alignment statistics that predict when training will behave like backprop (Sec. 5, Sec. 5.4). The directional sign-match $p > \frac{1}{2}$ emerges as a simple, testable criterion that links theory and measurement, helping to replace polarized narratives with cumulative evidence.

Practical impact. Ternary forwards reduce memory traffic and simplify integer datapaths while retaining float shadows for optimization. We quantify accuracy–sparsity–throughput trade-offs and CPU latency (Sec. 8.2, Sec. 8.3), yielding Pareto fronts that are directly actionable for edge deployments. In short, the method preserves core learning behavior while making deployability a first-class metric rather than an afterthought.

10 Limitations

Scope and ceilings. Our study targets compact benchmarks and shallow MLPs. The goal is a stability and deployability map under controlled budgets—not state-of-the-art accuracy. Bag-of-words text features cap ceiling performance; our emphasis is on statistically stable regimes and transparent comparisons (Sec. 6).

Architectural reach. While structured feedback improves conditioning, DFA/FA on deeper CNNs or Transformers remains fragile without additional design (e.g., hybrid or layer-wise feedback). Mixed outcomes on large convnets and ImageNet-scale tasks have been reported [16]; we intentionally avoid claims in those regimes.

Quantization and schedules. Thresholds that are too high stall learning; thresholds that are too low induce flip chatter. Per-step flips tighten coupling between shadows and ternary forwards but can destabilize some modalities; per-epoch flips trade reactivity for smoother gradients. These trade-offs are surfaced but not fully optimized here.

Theoretical coverage. Our bounds establish a descent-direction advantage and finite-time stationarity under assumptions (smoothness, $p > \frac{1}{2}$, bounded quantization variance). They do not provide global rates for deep nonlinear networks, nor guarantees in adversarial or heavy-tailed-gradient settings.

11 Future Work

Architectures. Extend FeedFlipNets to convolutional and attention-based models with feedback designs tailored to depth (e.g., block-local feedback, hybrid BP/DFA) and verify stability on medium-scale vision and language tasks.

Feedback design. Explore learned or adaptively orthogonalized feedback, closed-loop alignment (periodic reconditioning of B_ℓ), and fast transforms that preserve energy and improve gradient surrogates.

Quantization. Study per-layer thresholds, LSQ/DoReFa-style learnable quantizers, and joint gradient/activation quantization. Characterize how flip schedules interact with optimizer choice and gradient clipping.

Theory. Tighten finite-time rates under alignment assumptions, quantify depth dependence with structured feedback, and analyze noise shaping introduced by stochastic ternary thresholds.

Systems and tooling. Add export paths and kernels for microcontroller-class devices, integrate latency and throughput profilers into training runs, and provide lightweight harnesses for community reproduction and extension.

12 Conclusion

FeedFlipNets demonstrates that deterministic DFA with ternary forwards is both principled and practical. On the principled side, we isolate a measurable alignment statistic that predicts success, provide convergence guarantees under quantized updates, and clarify the role of structured feedback (Sec. 5). On the practical side, we deliver an end-to-end, fairness-minded artifact that reports accuracy, sparsity, throughput, and latency across modalities (Sec. 7, Sec. 8.2, Sec. 8.3).

Collectively, these components provide a transparent baseline and actionable guidance for when ternary-forward DFA succeeds and where it falls short. They invite rigorous follow-ons: scaling to richer architectures, refining feedback design and quantization, and closing the loop from reproducible research to deployable systems.

Appendix

A. Proof of Theorem 2

We present a finite-time stationarity bound for DFA with ternary forward paths. Recall $g_t = \nabla\mathcal{L}(V_t)$ and update direction $\tilde{g}_t$. Let $V_{t+1} = V_t - \eta\,\tilde{g}_t$ for constant $\eta \leq 1/(2L)$; the Robbins–Monro case follows by standard arguments.

Descent lemma. L-smoothness gives

$$\mathcal{L}(V_{t+1}) \leq \mathcal{L}(V_t) - \eta\,\langle g_t, \tilde{g}_t\rangle + \tfrac{L\eta^2}{2}\,\|\tilde{g}_t\|^2. \tag{7}$$

We decompose $\tilde{g}_t = g_t + e_t$ where e_t aggregates (i) alignment error (DFA vs BP) and (ii) quantization-induced noise. Taking expectations conditioned on V_t:

$$\mathbb{E}[\langle g_t, \tilde{g}_t\rangle] = \|g_t\|^2 + \mathbb{E}[\langle g_t, e_t\rangle] \geq (1-\alpha_t)\,\|g_t\|^2 - \beta_t, \tag{8}$$

for some $\alpha_t \in [0,1)$ and $\beta_t \geq 0$. The *sign advantage* $p > \frac{1}{2}$ ensures a nontrivial descent component, contributing a term $\propto (1-2p)^2$ to β_t. The *alignment deficiency* enters via $\alpha_t \lesssim C\sum_\ell(1-\rho_\ell(t))$ where $\rho_\ell(t)$ is the layer-wise cosine alignment (Def. 4).

Variance control. With stochastic ternary forward map Q_τ, the zero-mean jitter ε_t satisfies $\mathbb{E}[\varepsilon_t \mid V_t] = 0$ and $\mathbb{E}[\|\varepsilon_t\|^2 \mid V_t] \leq \sigma_q^2(\tau)$, implying

$$\mathbb{E}\|\tilde{g}_t\|^2 \leq 2\,\|g_t\|^2 + 2\,\mathbb{E}\|e_t\|^2 \leq 2\,\|g_t\|^2 + C'\,\sigma_q^2(\tau). \tag{9}$$

Telescoping. Taking expectations in (7) and substituting (8)–(9) yields

$$\mathbb{E}[\mathcal{L}(V_{t+1})] \leq \mathbb{E}[\mathcal{L}(V_t)] - \eta(1-\alpha_t-L\eta)\,\mathbb{E}\|g_t\|^2 + \tfrac{L\eta^2}{2}C'\sigma_q^2(\tau) + \eta\,\beta_t. \tag{10}$$

With $\eta \leq \frac{1}{2L}$ and $\alpha_t \in [0,1)$, the coefficient remains positive. Summing over $t = 1..T$ and lower bounding by $\mathcal{L}_\star$ telescopes and gives

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\|g_t\|^2 \leq \frac{2(\mathcal{L}(V_0)-\mathcal{L}_\star)}{\eta T} + C_1 L\eta\,\sigma_q^2(\tau) + C_2(1-2p)^2 + C_0\sum_\ell\big(1-\bar{\rho}_\ell(T)\big), \tag{11}$$

matching Eq. (4). The Robbins–Monro case follows by letting $\eta_t \to 0$ with $\sum \eta_t = \infty$, $\sum \eta_t^2 < \infty$.

B. Proof of Lemma 1

Assume B_ℓ row-orthonormal so that $\|B_\ell x\|_2 = \|x\|_2$. Let u_ℓ denote the backprop signal and v_ℓ the broadcast signal: $v_\ell = B_\ell \delta_L$ and $u_\ell = W_{\ell+1}^\top u_{\ell+1} \odot \phi'(z_\ell)$. The cosine alignment is $\rho_\ell = \langle u_\ell, v_\ell \rangle / (\|u_\ell\| \, \|v_\ell\|)$.

Because B_ℓ preserves norms and acts as a random rotation (orthonormal/Hadamard), the angle between u_ℓ and v_ℓ contracts in expectation under gradient descent on V_ℓ: the component of u_ℓ parallel to v_ℓ is reinforced, while the orthogonal component decays with a rate $e^{-\kappa}$ per step for some $\kappa \in (0, 1]$ depending on depth and layer Lipschitz constants. Averaging over T steps yields

$$\mathbb{E}\, \bar{\rho}_\ell(T) \geq \rho_\ell(0) e^{-\kappa T} + \gamma(1 - e^{-\kappa T}), \tag{12}$$

where γ is a depth-dependent floor induced by the near-isometry. For random orthonormal (or normalized Hadamard) B_ℓ, standard concentration provides $\gamma \gtrsim c/\sqrt{L}$ w.h.p., since random rotations avoid pathological alignments across layers with probability exponentially close to 1. This establishes the claim.

Dataset	E@0.90(ps)	S(ps)	E@0.90(pe)	S(pe)	dE	dS
MNIST	1	0	3	0.281	2	0.281
20NG	6	-0.008	6	0.078	0	0.086

Table 5: Schedule delta: per-step (ps) vs per-epoch (pe) flips. E@0.90: epoch to 90% Acc.; Slope: epoch 1 $\to$ epoch 3; Δ is pe $-$ ps.

Practical guidelines.

- **Feedback structure.** Prefer orthogonal or Hadamard B_ℓ for deeper stacks to preserve error magnitude and reduce tuning sensitivity.
- **Flip schedule.** Start with per-epoch flips for text or tabular data, then tighten to per-step flips on vision or time-series once training stabilizes.
- **Thresholding.** Sweep τ to target a zero ratio in the 0.5–0.7 range; push lower for accuracy, higher for compression.
- **Clipping and learning rate.** Clip gradients around 1.0 and tune learning rates conservatively under ternary forwards to avoid limit cycles.
- **Determinism.** Fix seeds for feedback and initialization to stabilize alignment onset and variance across runs.

Reproducibility Statement

We scope claims to shallow MLPs on small, standard datasets and provide all code, presets, and artifacts to reproduce results on a laptop-class machine.

Datasets and splits. MNIST, Fashion-MNIST (subsets), 20 Newsgroups, AG News (small variant), UCR GunPoint, and California Housing. We use canonical train/test splits; validation subsets are formed deterministically with fixed seeds.

Training protocol. Seeds are fixed for initialization and data shuffling. We run $n \in [2, 5]$ repeats per configuration and report means with 95% confidence intervals (Student's t) across seeds. We sweep learning rate and ternary threshold τ within specified ranges. No early stopping is used.

Implementation details. Feedback strategies include backprop, DFA (float), ternary-forward DFA, and structured feedback (orthonormal/Hadamard). Ternary quantization uses float shadow weights and deterministic or seeded stochastic rounding. Full configs are saved with each run.

Environment. CPU-only runs on a host with Apple M3 (Darwin 24.6.0, arm64). Dependencies are pinned via `requirements-lock.txt`. BLAS is provided via NumPy's linked backend on macOS; all latencies are measured with high-resolution timers on this host and reported alongside the device string in the Deployability table.

Artifacts. Per-run metrics/manifests under `runs/`; aggregate tables and plots under `data/report/`. Only figures backed by existing data are included via guarded includes; missing assets are omitted (no placeholders). The Key Metrics table is auto-generated for the paper build. Scripts: `reproduce_all.sh`, `make report`, and `make paper`.

Generating specific plots. To produce alignment and Pareto figures: enable alignment probes during training to log per-epoch sign-match and alignment (`metrics_train.jsonl`); use `scripts/plot_p_hist.py` to summarize directional sign-match p; and run `python scripts/make_pareto_plots.py` after threshold/schedule sweeps to generate accuracy–sparsity–throughput Pareto charts. These scripts read from `runs/` and write to `data/report/plots`.

Limitations. DFA/FA often underperform on deep CNNs and large-scale vision without adapted feedback; we do not claim results beyond our small MLPs. Partial sweeps (e.g., small n on some text datasets) are clearly marked; inferential claims rely on non-parametric paired tests and are treated as ties when not significant.

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